

Comprehensive Review of Calibration Techniques for Serial Link Manipulators: Insights into Mathematical and Methodological Evolutions

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Abstract—This paper reviews geometric calibration methods for serial link manipulators, focusing on improving their absolute positioning accuracy. The review highlights key calibration steps, including model selection, data acquisition, and parameter identification, and emphasizes the role of measurement and optimization. The review compares traditional geometric models with advanced non-minimal parameterizations, highlighting their respective advantages. It also examines modern data acquisition techniques using optical systems, fiducial markers, and SLAM, alongside optimization strategies for parameter refinement. The paper also outlines current challenges and future directions. This review supports researchers and practitioners in developing applications that require precise robot calibration.

Index Terms—Robot Calibration, Serial Link Manipulators, Geometric Modeling, Kinematic Analysis, Optimization Algorithms, Measurement Techniques

I. INTRODUCTION

A serial link manipulator consists of a fixed base, a series of links connected by motor-actuated joints, and ends with an end-effector. By actuating the joints, the end-effector can be positioned and oriented to perform a desired task. Serial manipulators are vital in both industrial and service robotics, where precise absolute positioning is increasingly critical for success. This need is intensified by advanced applications where trajectories are generated on-the-fly from real-time sensing, limiting reliance on pre-taught, fixed poses [1]–[4].

To describe and control the motion of serial manipulators, various kinematic and dynamic models are employed. These models typically rely on nominal parameters provided by the manufacturer; however, manufacturing tolerances, assembly errors, and mechanical wear introduce deviations that limit absolute positioning accuracy. Robot calibration addresses this issue by identifying robot-specific model parameters, thereby reducing systematic errors and improving pose accuracy in real-world applications.

These calibration procedures consist of several interconnected stages. First, an appropriate kinematic model is selected to formally describe the robot's motion based on its geometric structure. Next, during data acquisition, the end-effector pose is measured at multiple configurations using an external

measurement system, providing ground-truth observations. For each configuration, the expected end-effector pose is computed using the selected model and its nominal parameters. The discrepancy between the measured and model-predicted end-effector poses defines a set of pose residuals. These residuals are minimized through parameter identification, which iteratively refines the model parameters. The calibrated model is then subjected to a validation phase, during which its performance is evaluated against accuracy requirements. If these criteria are not met, the calibration process may be repeated by revisiting earlier stages such as model selection, data acquisition, or parameter optimization. An overview of the complete calibration workflow is presented in Fig. 1.

The primary objective of this paper is to provide a structured overview of the methodologies and algorithms employed in robot calibration, without performing direct experimental comparisons. By jointly addressing geometric modeling, measurement, and optimization aspects, the paper aims to serve as a practical reference for researchers and practitioners seeking calibration strategies for specific applications. In contrast to existing survey works [5]–[7], which mainly emphasize kinematic modeling and optimization techniques, this paper focuses on the often underrepresented measurement and data acquisition stage. As a key contribution, it introduces a structured taxonomy of measurement systems used in robot calibration (Table I), enabling a systematic comparison of sensing principles and information content.

The remainder of this paper is structured to systematically describe each stage of the calibration process illustrated in

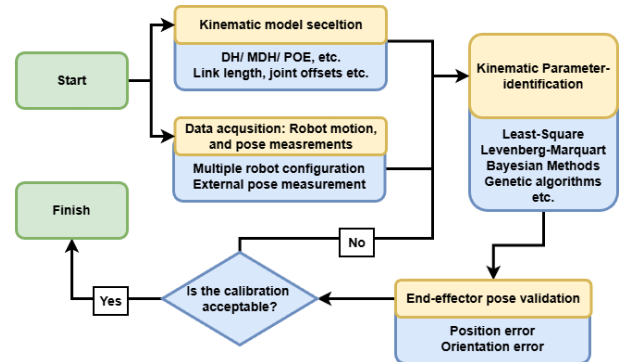


Fig. 1. Generic process of kinematic calibration of serial manipulators, with corresponding subprocesses, and examples.

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Fig. 1. For each stage, concise methodological explanations are provided along with key practical considerations. Section II reviews the historical development and mathematical foundations of robot modeling. Section III introduces geometric modeling techniques, while Section IV presents and categorizes the measurement systems commonly used in calibration procedures. Parameter identification and optimization methods are discussed in Section V. Section VI introduces a proposed comparison metric for robot calibration approaches, addressing the current lack of a standardized method for objective evaluation. Section VII outlines open challenges and emerging research directions in the field. Finally, Section VIII concludes the paper.

II. HISTORICAL PERSPECTIVES

The mathematical foundations of robot calibration can be traced back to the development of linear algebra and analytical geometry. René Descartes' formulation of coordinate geometry in 1637 enabled the systematic representation of geometric relationships through systems of linear equations. This framework was later strengthened by Gottfried Wilhelm Leibniz, who introduced determinant-based methods for solving linear systems in 1693.

Building on these developments, the representation of rotational transformations in a compact form was introduced independently by Olinde Rodrigues in 1816, Sir James Ivory in 1824, and Carl Gustav Jacobi in 1827, and is now known as the Rodrigues rotation formula [8]. This formula provided an efficient and concise representation of three-dimensional rotations, which later became a fundamental tool in kinematic analysis.

In 1830, Michel Chasles further advanced the understanding of rigid-body motion by proving that any spatial displacement can be expressed as a combination of rotation about an axis and translation along the same axis [9]. This result, known as the Mozzi–Chasles theorem, remains a cornerstone of modern robotic modeling.

Based on these foundational principles, two primary approaches for describing rigid-body transformations emerged, both rooted in Chasles' theorem but employing complex mathematical formalisms involving six or more parameters. One such approach is the dual-quaternion representation, which originated from the work of William Rowan Hamilton [10], allowing for the representation of rotational transformations. This was later extended by W. K. Clifford in 1873 [11] to encompass general spatial displacements. The practical application of dual-quaternions was further established by Eduard Study in 1901 [12], and these methods were later integrated into robot modeling and calibration techniques starting in the 2000s [13], [14].

Alongside dual-quaternions, another significant modeling technique emerged. In 1865, Julius Plücker introduced Plücker coordinates [15], which were subsequently extended to represent screws and wrenches in robot kinematics, as demonstrated by Sir Robert Stawell Ball in 1876 [16]. This work was based on the earlier contributions of Sophus Lie, who laid the foundations for Lie group theory between 1873 and

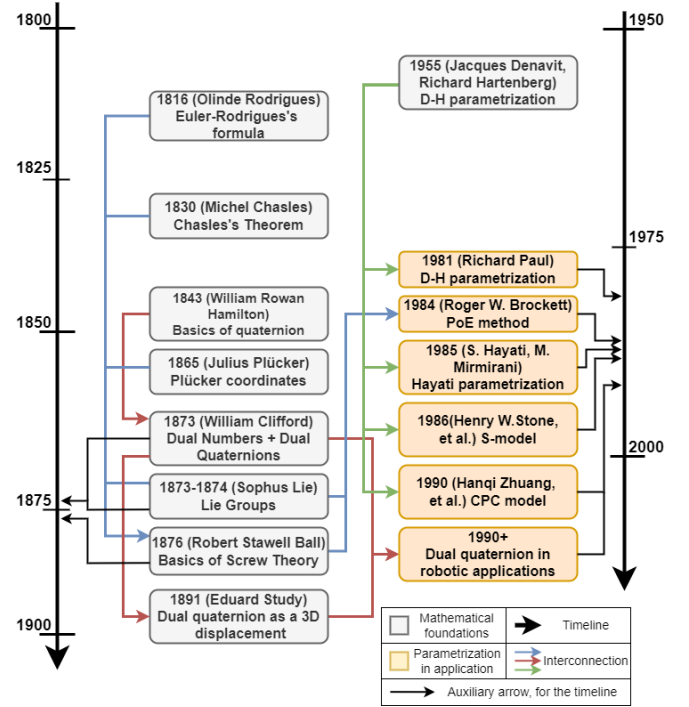


Fig. 2. Historical development of robot parameterization and calibration techniques.

1874 [17]. These advancements eventually led to the product of exponentials formulation introduced by Roger W. Brockett in 1984 [18], which has since been widely applied in robot calibration [19]–[22].

In the 1950s, Jacques Denavit and Richard Hartenberg introduced a new development aimed at standardizing coordinate frames in a compact way for spatial linkages [23]. Their approach played a critical role in kinematic analysis, with further contributions by Richard Paul in 1981 [24]. This standardized framework has influenced several subsequent models, such as the Hayati-Mirmirani method (1985) [25], the Stone S-model (1986) [26], and the Complete and Parametrically Continuous model (1990) [27], which sought to address the limitations of earlier methods while maintaining the standardization.

The key developments discussed above are summarized in the historical timeline presented in Fig. 2, which illustrates the interconnections between these approaches and their progression into practical applications.

III. GEOMETRIC MODELING TECHNIQUES FOR SERIAL MANIPULATORS

Geometric modeling provides a formal mathematical description of serial manipulators and defines the parameter space required for calibration. As illustrated in Fig. 1, the choice of modeling technique is critical, as it directly influences the formulation and solution of the calibration problem. This section reviews modeling approaches that explicitly relate joint variables to the pose of the end-effector with respect to the robot base.

Most classical calibration methods rely on explicit geometric models in which coordinate frames are assigned to individ-

ual links. The spatial relationship between consecutive links is described by six independent parameters, three translational and three rotational, and the pose of the tool center point (TCP) is obtained by sequentially composing these transformations along the kinematic chain:

$$\mathbf{T}_{\text{TCP}}^{\text{Base}} = \mathbf{T}_0^{\text{Base}} \mathbf{T}_1^0 \mathbf{T}_2^1 \dots \mathbf{T}_n^{(n-1)} \mathbf{T}_{\text{TCP}}^n, \quad (1)$$

where $\mathbf{T}_k^i$ denotes the homogeneous transformation between the i -th and k -th coordinate frames.

In contrast, model-free approaches dispense with explicit geometric modeling and instead treat the manipulator as a black-box system, approximating its behavior through soft computing techniques such as neural networks [28]–[31], and fuzzy models [32]. Although these methods offer high flexibility, they typically require large training datasets and lack physical interpretability. For these reasons, such approaches are not considered within the scope of this review.

The following subsections review common geometric parameterizations and their advantages and limitations.

A. Ad-hoc approaches

This approach does not provide a generalized formulation that directly relates consecutive links; instead, each frame pose is defined relative to the base by accumulating all preceding transformations. A representative example is the Unified Robot Description Format (URDF) used in the Robot Operating System (ROS) [33]. Although URDF is widely adopted and well suited for simulation, it is not directly applicable to analytical modeling or control, motivating conversion methods to computational forms such as the Denavit–Hartenberg parameterization [34], [35].

B. Minimal parameter representations

As noted previously, at least six independent parameters are required to clearly define the transformation between consecutive links. Minimal parameter representations reduce the number of parameters for each consecutive transformation, simplifying the modeling process. However, this reduction relies on strict reference frame assignment rules, which, while advantageous to the complexity, introduce limitations, particularly in the form of singularity issues. The following subsections introduce various minimal parameter representations and examine their respective advantages and limitations.

1) *Denavit–Hartenberg (DH) Representation*: The Denavit–Hartenberg (DH) parameterization is the most widely adopted geometric modeling approach for serial robotic manipulators [23], [36]. This formulation describes the spatial relationship between consecutive link frames using four parameters. The homogeneous transformation between the $(i-1)$ -th and i -th frames is defined as:

$$\mathbf{T}_i^{(i-1)} = \text{Tr}_z(d_{(i-1)}) \text{Rot}_z(\theta_{(i-1)}) \text{Tr}_x(a_i) \text{Rot}_x(\alpha_i), \quad (2)$$

where $\mathbf{T}_i^{(i-1)}$ denotes the transformation between consecutive frames, $\text{Tr}_a(b)$ represents a translation of magnitude b along the axis a , and $\text{Rot}_a(b)$ denotes a rotation of angle b about the axis a .

The overall transformation from the robot base to the TCP is obtained by sequentially multiplying the individual link transformations, as expressed in Equation 1. As the DH convention is a minimal parameter representation, it inherently introduces singularities. In particular, when consecutive joint axes are parallel or nearly parallel, the offset parameter d becomes ill-defined. These singular configurations can negatively affect parameter identification. Several studies have therefore investigated strategies for robust identification of DH parameters [37]–[39] for serial manipulators.

2) *Modified Denavit–Hartenberg (MDH) Representation*: The DH parameterization can yield unnecessarily complex models for manipulators with intricate structures. To address this, the Modified Denavit–Hartenberg (MDH) parameterization was introduced [40].

MDH employs the same four geometric parameters as the standard DH convention but differs in reference frame attachment and transformation order. While DH assigns the i -th frame to the output side of the $(i-1)$ -th link, MDH places it on the input side of the link, as discussed in [41]. Although this modification can simplify certain kinematic descriptions, it does not eliminate the singularities inherent to minimal parameterizations, and parameters such as d may remain ill-defined when consecutive joint axes are parallel. The resulting MDH transformation sequence is defined as follows:

$$\mathbf{T}_i^{(i-1)} = \text{Rot}_x(\alpha_{(i-1)}) \text{Tr}_x(a_{(i-1)}) \text{Rot}_z(\theta_i) \text{Tr}_z(d_i), \quad (3)$$

where the transformations are denoted consistently with the notation introduced in Equation 2, and the overall transformation from the base to the TCP is computed as defined in Equation 1.

3) *Hayati–Mirmirani Representation*: To address the singularities that arise in the DH and MDH conventions when consecutive joint axes are parallel, Hayati and Mirmirani introduced an alternative geometric parameterization [25]. This parameterization modifies the ordering of the elementary transformations associated with the parameters in the definition of the transformation between consecutive links. This modification results in a singularity when consecutive joint axes are perpendicular or nearly perpendicular [42], [43]. Consequently, the transformation between consecutive links is defined as:

$$\mathbf{T}_i^{(i-1)} = \text{Rot}_z(\theta_i) \text{Tr}_x(a_i) \text{Rot}_x(\alpha_i) \text{Rot}_y(\beta_i), \quad (4)$$

where the transformations are denoted consistently with the notation introduced in Equation 2, and the overall transformation from the base to the TCP is computed as defined in Equation 1.

4) *Combined Minimal Representation*: To mitigate the singularity-related limitations inherent to minimal parameter representations, hybrid modeling strategies have been proposed that combine multiple minimal representations by exploiting their complementary properties. In practice, the DH parameterization is typically employed; however, when consecutive joint axes are parallel or nearly parallel, the Hayati–Mirmirani parameterization is adopted to avoid the associated singularities [44], [45]. Explicit conversion procedures between the DH and Hayati–Mirmirani parameter sets

are provided in [46], enabling seamless transitions between the two representations for more efficient handling.

C. Non-minimal representations

To avoid the singularities inherent to minimal parameterizations without employing hybrid models, non-minimal representations have been introduced for robot geometric modeling [47]. These approaches use at least six parameters per joint to provide a unique and continuous description of the end-effector pose.

Although non-minimal models are typically more abstract and computationally demanding than minimal representations, they offer parameter continuity and improved robustness. The following subsections summarize commonly used non-minimal parameterizations along with their advantages and limitations.

1) *Stone (S)-model*: Stone extended the DH parameterization by introducing two additional parameters, resulting in the S-model [26]. Unlike the standard DH convention, the S-model allows the origin of the i -th frame to be placed arbitrarily along the $(i + 1)$ -th joint axis and permits free selection of the x -axis orientation. These additional degrees of freedom enable infinitely many valid frame assignments while preserving a compact parameterization comparable to DH. Consequently, the S-model avoids singularities and provides increased flexibility, making it well suited for robot calibration tasks.

The homogeneous transformation between consecutive coordinate frames in the S-model is defined as:

$$\mathbf{T}_i^{(i-1)} = \text{Rot}_z(\beta_i) \text{Tr}_z(d_i) \text{Tr}_x(a_i) \text{Rot}_x(\alpha_i) \text{Rot}_z(\gamma_i) \text{Tr}_z(b_i), \quad (5)$$

where the transformations are denoted consistently with the notation introduced in Equation 2, and the overall transformation from the base to the TCP is computed as defined in Equation 1.

2) *Complete and Parametrically Continuous model (CPC)*: The CPC model extends the classical Denavit–Hartenberg formulation by introducing two additional parameters and by separating joint-dependent and joint-independent components of the kinematic transformation [27]. This design yields a complete and continuous representation of robot geometry, allowing arbitrary geometric deviations to be modeled while avoiding singularities [48]. Owing to these properties, the CPC model is used in robot calibration and geometric parameter identification [49]–[51].

The transformation between consecutive links is expressed as

$$\mathbf{T}_i^{(i-1)} = \mathbf{Q}_i \mathbf{V}_i = \mathbf{Q}_i \mathbf{R}_i \text{Rot}_z(\beta_i) \text{Tr}(l_{ix}, l_{iy}, l_{iz}), \quad (6)$$

where $\mathbf{Q}_i$ denotes the joint-motion transformation, $\mathbf{R}_i$ aligns successive joint axes, $\text{Rot}_z(\beta_i)$ is a rotation about the z -axis, and $\text{Tr}(\cdot)$ represents translation along the Cartesian axes by l_{ix}, l_{iy}, l_{iz} values.

3) *Product of Exponentials (PoE) Formulation*: The Product of Exponentials (PoE) formulation offers several advantages, including a strong theoretical foundation in Lie group theory [17], inherent avoidance of kinematic singularities, and the ability to determine joint parameters independently without

requiring propagation through preceding joints. The mathematical formalism of PoE is summarized in [52]; however, since the focus of this work is on reviewing modeling approaches rather than their mathematical derivation, only the essential concepts are summarized here.

In the PoE framework, rigid-body motion is represented as a product of exponentials of screw motions. The forward kinematics of a serial manipulator can be expressed as

$$\mathbf{T}_{\text{TCP}}^{\text{Base}}(\boldsymbol{\theta}) = \left(\prod_{i=1}^m e^{[\mathcal{S}_i]\theta_i} \right) \mathbf{T}_{\text{TCP}}^{\text{Base}}(\mathbf{0}), \quad (7)$$

where $\mathbf{T}_{\text{TCP}}^{\text{Base}}(\boldsymbol{\theta})$ denotes the end-effector pose for joint configuration $\boldsymbol{\theta}$, and $\mathbf{T}_{\text{TCP}}^{\text{Base}}(\mathbf{0})$ corresponds to the zero-reference configuration. Each term $e^{[\mathcal{S}_i]\theta_i}$ represents the homogeneous transformation induced by the i -th joint, computed via the matrix exponential of the associated screw vector $\mathcal{S}_i \in \mathbb{R}^6$ scaled by the joint variable θ_i .

4) *Dual quaternions*: Dual quaternions are widely used in computer graphics and animation [53], [54] and have gained increasing attention in robotics for modeling rigid-body motion [55], [56]. Their formulation builds upon the concepts of dual numbers and quaternions.

- **Dual numbers**: Dual numbers were introduced by Clifford in 1873 [11] as an extension analogous to complex numbers. A dual number is defined as

$$z = r + d\varepsilon, \quad \varepsilon^2 = 0, \quad \varepsilon \neq 0, \quad (8)$$

where r denotes the real part, d the dual part, and ε is the dual unit. Similar to the imaginary unit in complex numbers, ε enables the separation of real and dual components. Dual numbers have been extensively applied in kinematics and robotics, particularly for representing translational quantities [57], [58]. In this work, their role is considered primarily in combination with quaternions.

- **Quaternions**: Quaternions were introduced by Hamilton in 1843 [10] as a four-dimensional extension of complex numbers. They provide a compact and singularity-free representation of three-dimensional rotations and are widely used in robotic modeling and motion representation. For a detailed introduction to quaternion algebra and its applications in robotics, the reader is referred to [59]–[61].
- **Dual quaternions**: By combining dual numbers with quaternions, Clifford introduced dual quaternions in 1873 [11]. While a unit quaternion represents pure rotation, a unit dual quaternion encodes both rotation and translation in a unified algebraic structure. This makes dual quaternions particularly suitable for robot geometric modeling, as they inherit key advantages of quaternion algebra. The presentation of dual quaternions in this work is based on the formulations provided in [62], [63].

IV. MEASUREMENT APPROACHES IN ROBOT CALIBRATION

The second phase of robot calibration is data acquisition (Fig. 1), whose objective is to collect a sufficient amount

		Measurement principles		
		Mechanical constraints	Distance measurement	Distance + direction measurement
DoF considered in one elementary measurement	1 DoF	-	Single draw-wire encoder: [66]–[69]	-
	3 DoF	Point and line constraints: [70]–[72] Surface constraints: [73]–[80]	Multiple draw-wire encoder: [81], [82]	Sphere localization: [83], [84] Fiducial markers: [85]–[88] Laser tracker: [89]–[94] SLAM approaches: [95], [96]
	6 DoF	Closed loop using two robots: [97]	Parallel draw-wire mechanism: [98]–[100] Six-Point measuring device: [101]	Sphere localization: [102] Fiducial markers: [14], [103]–[109] Laser tracker: [110], [111]

TABLE I
OVERVIEW OF MEASUREMENT APPROACHES DERIVED FROM THE ANALYZED LITERATURE.

of accurate data linking joint coordinates to Cartesian end-effector poses [64], [65]. Despite its importance, data acquisition strategies are less frequently studied than modeling and optimization methods.

In this phase, Cartesian measurements are obtained using external sensing systems operating independently of the robot, while joint values are read from the controller. The available pose information varies by method, ranging from mechanically constrained setups aligned to geometric primitives to measurement systems capturing one to six scalar values. These approaches provide different levels of information for parameter identification, depending on their sensing principle and measurement precision.

Measurement systems can therefore be categorized by the number of degrees of freedom they measure and their underlying sensing technology. An overview of these categories is summarized in Table I, and the following section reviews them in detail.

A. Mechanical constraints

This approach relies on mechanical fixtures to constrain the end-effector to predefined points or lines in Cartesian space. While it is conceptually simple and requires only a small number of precisely identified geometric features, achieving high accuracy across the entire workspace necessitates a well-distributed set of constraints. As a result, scalability becomes a limiting factor for global calibration. The following examples illustrate representative implementations of this approach, including methodological details and the achieved positioning accuracy [70]–[78].

1) *Point and line constraints*: This approach uses mechanical fixtures to constrain the end-effector to predefined points or lines in Cartesian space. While simple and requiring only a

limited number of precisely identified features, achieving high accuracy over the full workspace demands a well-distributed set of constraints, which limits scalability for global calibration.

Lee et al. [70] proposed a calibration method in which the end-effector position is mechanically constrained while its orientation is varied by contacting a fixed workpiece at multiple locations. From the recorded robot configurations, theoretical end-effector positions were derived, achieving a reduction in positioning error from 5.7 mm to 0.885 mm using three workpiece locations and 36 poses each, validated over 2000 random configurations.

Similarly, Veitschegger and Wu [71] employed a peg tool inserted into fixed holes, exploiting relative distance constraints rather than absolute positioning. Using 40 hole locations, the method reduced the positioning error from 6–8 mm to below 1 mm.

Liu et al. [72] calibrated a neurosurgical robot by constraining the end-effector to follow linear and circular trajectories while measuring its position with a FARO arm. With 19 configurations, the mean absolute positioning error was reduced from 3.7 mm to 0.7 mm.

2) *Surface constraints*: This approach constrains the end-effector using fixtures that define continuous geometric surfaces, such as planes or spheres. By moving the end-effector along these surfaces across multiple configurations, measurement data are obtained for calibration.

Although this method increases measurement complexity, since the constrained surfaces or their geometric centers must be determined with high accuracy, it offers improved global calibration performance. Unlike point- or line-based constraints, continuous surface constraints provide information over extended regions of the workspace, resulting in more consistent positioning accuracy. Representative implementations

are reviewed below.

Planar constraints have been employed in several studies [73]–[75], where the end-effector contacts or moves along calibrated planes using a touch probe. These methods report reductions in absolute positioning error from 10–16 mm to approximately 1–1.5 mm with around 120 measurement samples. Zhuang et al. [73] showed that a single plane provides insufficient information, while Besnard et al. [74] demonstrated improved results when using multiple planes.

Spherical constraints have also been investigated [77], [78]. Driels et al. [77] used a ball-bar mechanism to constrain the motion of the robot to a spherical surface, achieving a mean positioning error of 0.3 mm using 44 configurations. Joubair and Bonev [78] employed a multiple probe-based contact strategy, and three spheres affixed to a rigid base. Theoretical sphere centers were computed from joint data and compared with actual distances. Using 30 poses per sphere, calibration reduced error from 1.321 mm to 0.127 mm.

3) *Closed chain using two robots*: This calibration method forms a closed kinematic chain using two serial manipulators: one accurately calibrated reference robot and one uncalibrated robot. The manipulators are mechanically coupled at their bases and end-effectors, and joint data are recorded while the system moves through multiple configurations. Relative pose discrepancies are then used to identify and correct the geometric parameters of the uncalibrated robot [97].

The approach is simple and requires no external measurement devices; however, it depends on the availability of a highly accurate reference robot and a rigid, precisely manufactured coupling, which limits its practicality.

B. Distance measurement

These calibration approaches employ distance-measuring sensors or encoders, most commonly draw-wire encoders or linear displacement sensors mounted on the robot end-effector. By moving the robot through a set of configurations and recording the corresponding distance measurements, geometric parameters can be estimated through calibration.

The principal advantage of distance-based sensing lies in its simplicity, as both the hardware and the acquired measurements are limited to scalar distance values. However, relying solely on such scalar information often leads to a mathematically challenging calibration problem. Furthermore, estimating multiple degrees of freedom, such as full Cartesian position or orientation, requires multiple sensors and increasingly complex measurement setups, which limits the scalability of these methods for higher-dimensional pose estimation. The following examples illustrate representative implementations and their achieved accuracy [66]–[69], [81], [82], [98]–[101].

1) *Draw-wire encoder-based approaches*: Draw-wire encoders measure the length of an extended cable and are commonly used for distance-based robot calibration. In their simplest form, a single encoder is mounted at a fixed location, with the wire attached to the robot end-effector. As the robot moves through multiple configurations, while avoiding wire bending, the measured cable lengths and corresponding joint values are used for calibration optimization. One example is

provided by Boschetti and Sinico [69] used 70 measurement poses and a single draw-wire encoder and showed a reduction in the mean positioning error from approximately 2 mm to 0.3 mm.

To capture additional degrees of freedom within a single measurement, multiple draw-wire encoders can be employed. In [81], [82], four encoders were used to capture the position of the end-effector within a single measurement. Using 30 and 87 poses, these studies reduced mean positioning errors from 4 mm to 1.3 mm and from 12 mm to 1.2 mm, respectively.

An extension of this approach employs six draw-wire encoders mounted on a fixed base to measure the full 6D pose of a rigid platform attached to the robot flange. The resulting measurement configuration is analogous to a Stewart platform [112], with draw-wire encoders replacing linear actuators to provide length measurements. Using this setup, Sakai et al. [100] collected between 50 and 102 measurement poses and demonstrated a reduction in mean positioning error from approximately 6.6–6.75 mm to 1.4–1.7 mm.

2) *Six-Point measuring device*: Ying Liu et al. [101] presented a method that used a measurement device equipped with six displacement sensors to measure the pose of a gauge block mounted on the robot flange. The gauge block was inserted into the measurement device at multiple robot configurations. In the experimental evaluation, 35 robot configurations were used for calibration, with an additional 9 configurations reserved for validation. The proposed method showed a mean absolute positioning error reduction from 3 mm to 0.1 mm and the orientation error from 0.5° to 0.1° . Furthermore, a coordinate measuring machine validation showed a reduction in the maximum positioning error from 3.8 mm to 0.5 mm.

C. Distance and direction measurement

These calibration approaches use sensors that measure both the distance and direction of observed objects or markers relative to the sensor frame, typically employing optical systems such as cameras or laser trackers. Two common configurations are used: eye-in-hand, where the sensor is mounted on the robot end-effector and observes a fixed object, and eye-to-hand, where the sensor is fixed in the environment and observes a marker attached to the robot.

A major advantage of these methods is their simplicity, as calibration can be performed using a single sensor and a known object, without complex measurement setups, and the measured poses can be used directly for optimization. However, measurement accuracy depends on external factors, including sensor–object distance and, for camera-based systems, illumination conditions, surface properties, and image quality, which must be carefully considered in practice. The following subsections review representative calibration methods based on this principle [14], [83]–[96], [102]–[108], [110], [111].

1) *Sphere localization*: These calibration methods exploit vision-based localization of spherical reference objects, whose centers can be robustly detected in camera images. In [83], [84], [102], cameras mounted on the robot end-effector observe fixed spheres while the robot moves through multiple configurations. Since the sphere positions remain constant,

deviations in the estimated sphere centers directly reveal positioning errors, enabling geometric parameter identification. Using this principle, Švaco et al. [84] reduced the mean absolute positioning error from 3.6 mm to 1.3 mm with 24 calibration poses, while Yang et al. [102] achieved a reduction from 8.423 mm to 2.864 mm using an RGB-D camera and 45 configurations.

2) *Fiducial markers*: These methods use fiducial markers such as checkerboards [14], [85], [86], [105], [106], ArUco markers [87], [107], [108], or laser points [88].

Yan Meng and Hanqi Zhuang [103] applied an eye-in-hand checkerboard-based method with 15 robot configurations, reducing the absolute positioning error from 5 mm to 0.369 mm.

Two studies [14], [106] explored dual-robot calibration, with one robot carrying the camera and the other holding the marker. While one robot remained stationary in various configurations, the other captured marker images from multiple poses. As described in Section IV-A3, if one robot is well-calibrated, it can serve as a reference to calibrate the other. This approach has been shown to significantly reduce the misalignment error between the two robots. This method reduced misalignment errors from 2.22 mm to 0.33 mm in the first, and from 2.28 mm to 0.25 mm in the second study.

Park, In-Won et al. [88] proposed a method using a structured laser module (SLM) on the robot's end-effector, a fixed camera, and a screen. As the robot moved to several configurations, the SLM projected laser dots onto the screen, which were detected in the camera images. Using known camera parameters and the positions of the laser points on both the camera image and the screen, the robot could be accurately calibrated.

3) *Laser tracker*: These calibration approaches adopt an eye-to-hand configuration, in which a laser tracker remains fixed in the environment while one or more reflector targets are mounted on the robot end-effector. By arranging multiple reflectors, a reference coordinate frame can be established for accurate 3D or 6D pose measurement, as demonstrated by Nubiola and Bonev [110].

Several studies report substantial accuracy improvements using this principle. In [89], [110], 200 and 52 calibration poses were used, reducing mean positioning errors from 4.07 mm to 0.81 mm and from 0.97 mm to 0.36 mm, respectively. Chen et al. [90] further analyzed the influence of workspace coverage by comparing two calibration datasets of equal size (50 poses) spanning different volumes. Calibration over the larger workspace ($450 \times 450 \times 450 \text{ mm}^3$) achieved a lower mean error (0.34 mm) than the smaller volume ($300 \times 300 \times 300 \text{ mm}^3$, 0.46 mm), with both models validated across the full workspace.

4) *SLAM approaches*: These approaches integrate Simultaneous Localization and Mapping (SLAM) with an RGB-D camera mounted on the robot end-effector. As the robot moves through its workspace, the SLAM algorithm incrementally reconstructs a 3D map of the environment while simultaneously estimating the camera pose by tracking visual features. By recording the robot joint parameters together with the corresponding camera poses across multiple configurations, a dataset is obtained that can be used for calibration. Jinghui Li

et al. [96] employed this strategy using 73 images, achieving calibration results that were validated with a laser displacement sensor. Their method reduced the maximum absolute positioning error from 12 mm to 3.9 mm.

V. PARAMETER-IDENTIFICATION AND OPTIMIZATION METHODS

Once the robot model and measurement data are available, a parameter identification step is performed (Fig. 1). This step evaluates the discrepancy between poses calculated by the nominal model and measured ground-truth poses, defining a typically nonlinear, non-convex optimization problem. Calibration aims to minimize this error by adjusting the model parameters. Various optimization strategies have been proposed for this purpose; representative approaches are reviewed in the following subsections.

A. Least-Squares Algorithm

Least-squares methods [113] estimate model parameters by minimizing the sum of squared residuals between measured data and model predictions. These methods are classified as linear or nonlinear, with nonlinear least-squares being used in robot kinematic calibration. While computationally efficient and easy to implement, they are sensitive to initial parameter estimates and can converge to local minima rather than the global optimum.

Similarly, Albert Nubiola and Ilian A. Bonev [79] employed a nonlinear least-squares algorithm to calibrate a Fanuc LR Mate 200iC robot using a 25-parameter Modified Denavit–Hartenberg (MDH) model. Their nonlinear least-squares methodology reduced the average/maximum positioning error from 1.355 mm/5.197 mm to 0.769 mm/1.630 mm.

B. Levenberg-Marquardt Algorithm

The Levenberg–Marquardt algorithm (LMA), or damped least-squares, is a method for solving nonlinear least squares problems. It combines elements of the Gauss–Newton algorithm (GNA) [114] and gradient descent [115]. The LMA is more robust than GNA, finding solutions even when initial parameters are far from the optimum, though it is slower for well-behaved functions with good initial guesses.

This optimization technique is extensively employed in robot calibration [103]–[105]. Du Guanglong and Zhang Ping [105] proposed an eye-in-hand calibration approach in which a combined kinematic modeling strategy was adopted, while the LMA was used for optimization. In their experimental evaluation, the robot operated under payloads of 0.5 kg, 0.8 kg, and 1.2 kg, and seven pose measurements were collected for parameter identification. The prior mean absolute positioning errors were 5.62 mm, 6.85 mm, and 7.33 mm, with corresponding standard deviations of 0.61 mm, 0.72 mm, and 0.81 mm. These mean absolute errors were reduced to 1.22 mm, 1.34 mm, and 1.53 mm, respectively after calibration.

C. Bayesian Estimation Methods

Bayesian estimation techniques, such as Kalman filters and Particle filters, are widely employed in robotics for state estimation, which is also applicable to robot calibration. The Kalman filter and its extensions, including the Extended Kalman Filter (EKF), fuse prior model predictions with measurement updates to improve estimation accuracy. While the standard Kalman filter is limited to linear system models, the EKF addresses nonlinear systems through local linearization, making it suitable for the inherently nonlinear and non-convex calibration problem. Comprehensive introductions to these methods can be found in [116]–[118].

Several studies have applied advanced filtering techniques, particularly the Extended Kalman Filter (EKF), to robot calibration [89], [93], [109]. Nguyen et al. [89] used a combined minimal model together with an EKF and artificial neural network (ANN) calibration to compensate for nonlinear joint effects. Their method was tested in both simulation and real environments, and it was validated by a laser tracker system. They achieved a reduction in average/maximum positioning errors from 8.35/15.90 mm to 0.04/0.08 mm in simulation and from 3.59/6.15 mm to 0.42/0.92 mm on a real Hyundai HH800 robot.

D. Maximum Likelihood Estimation

Maximum likelihood estimation (MLE) is a statistical approach for estimating model parameters by selecting the values that make the observed data most probable under an assumed probabilistic model [119]. In robot calibration, MLE is used to identify geometric parameters that best explain the measured pose data while accounting for measurement uncertainty. Due to its statistical rigor and flexibility in handling noise, MLE has become a widely adopted parameter estimation technique in calibration and related inference problems. This method has been utilized in several studies, including those described in [92], [111].

An example of this optimization strategy is presented by Ma Le et al. [92], who proposed a robot calibration method based on an extended geometric model that includes both joint-dependent and joint-independent parameters. During data acquisition, a laser tracker was used to measure the three-dimensional Cartesian coordinates of the robot's end-effector. The proposed approach led to a substantial improvement in absolute positioning accuracy, reducing the mean error from 8.721 mm to 0.277 mm.

E. Nelder–Mead Method

The Nelder–Mead algorithm [120] is a derivative-free direct search optimization method that has been applied in several robot calibration studies [70], [80], [96]. It is designed to locate extrema of objective functions in multidimensional spaces and is particularly suitable for nonlinear optimization problems where gradient information is unavailable or unreliable. However, due to its heuristic nature, the method does not guarantee convergence to a true stationary point and may, in some cases, converge to suboptimal or non-stationary solutions.

An example of this optimization approach is provided by Nadeau et al. [80], who proposed a robot calibration method based on spherical geometric constraints, using three spheres at known locations contacted by a trihedral kinematic coupling mounted on the robot end-effector. Joint configurations were recorded while minimizing deviations between measured and theoretical inter-sphere distances using the Nelder–Mead algorithm. Validation was performed using a FARO ION laser tracker over 125 configurations, reducing the mean positioning error from 1.403 mm to 0.677 mm.

F. Genetic Algorithm

Genetic algorithms (GAs) are population-based optimization techniques inspired by evolutionary principles [121]. In this framework, each individual represents a candidate solution encoded as a parameter vector, whose quality is assessed through a fitness function.

The optimization process starts with a randomly initialized population and iteratively evolves it using selection, crossover, and mutation operators, progressively favoring individuals with higher fitness. Owing to their global search capability and robustness against local minima, GAs have been widely applied to robot calibration problems [94], [122], [123]. Despite these advantages, GA-based methods are typically computationally intensive. Their execution time strongly depends on the population size, the number of generations, and the cost of evaluating the fitness function, which can limit their practicality for large-scale or real-time calibration tasks. Further methodological details can be found in [124], [125].

Liu Yu et al. [123] proposed a simulation-based robot calibration approach employing a genetic algorithm for parameter identification. Their method is built on a geometric error model formulated using Denavit–Hartenberg (DH) parameters and explicitly accounts for measurement system inaccuracies by modeling them as normally distributed noise variables. End-effector poses were sampled in multiple non-singular robot configurations, and the calibration parameters were optimized using a GA-based strategy. Simulation results demonstrated a significant improvement in the robot's absolute positioning accuracy.

VI. COMPARISON AND CALIBRATION EFFICIENCY INDEX

Calibration methods vary widely in terms of modeling, measurement systems, and optimization techniques, and no single method is universally optimal for all scenarios.

Measurement systems come with their own trade-offs: Mechanical constraints are simple, cost-effective, and easy to implement but lack scalability. Distance-based methods are similarly easy to implement but provide limited data. On the other hand, vision- and laser-based systems offer greater flexibility and allow for direct pose estimation, though their performance is heavily influenced by sensor accuracy and environmental conditions.

Optimization methods also have distinct strengths and weaknesses. Gradient-based methods provide a good balance between accuracy and computational efficiency, though they can be prone to local convergence. Probabilistic methods enhance

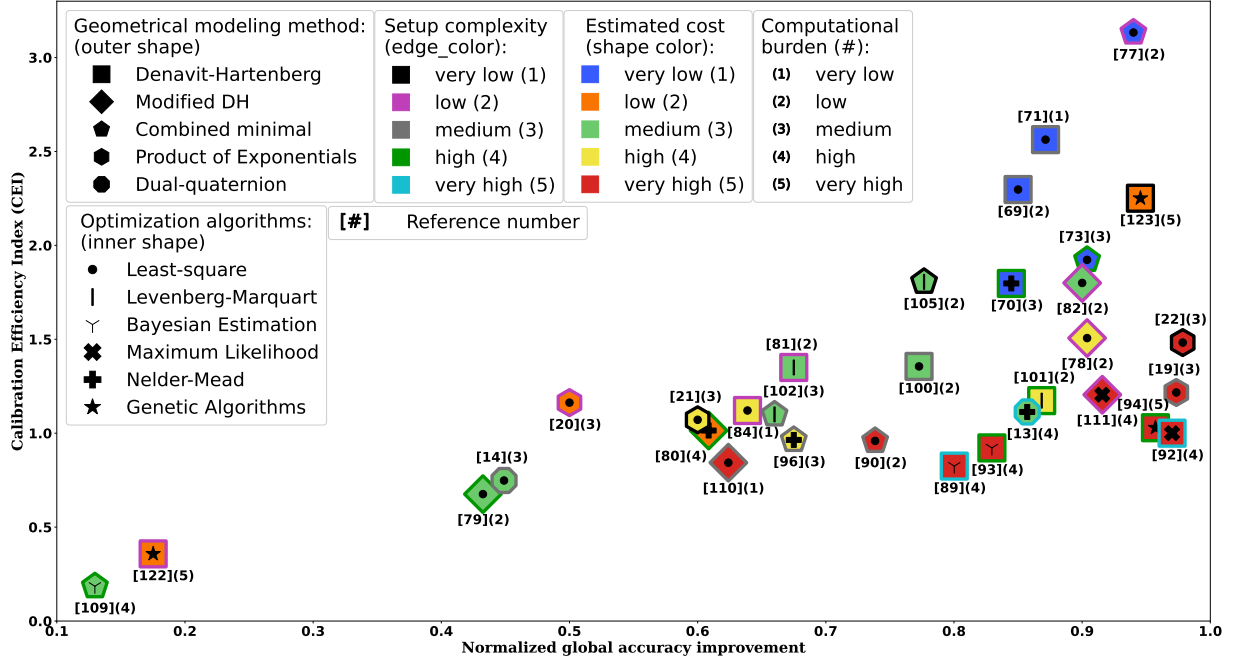


Fig. 3. Comparison of different calibration approaches considering the accuracy improvement, the CEI metrics, the estimated cost, the robot modeling approach, the optimization algorithms, the setup complexity, and the computational cost.

robustness to noise and support adaptive calibration, but they come with increased complexity. Global optimization techniques, while more comprehensive in exploring the parameter space, are computationally demanding.

These variations illustrate why objective comparison is challenging, as results are often influenced by additional factors such as robot type, workspace size, measurement density, calibration setup, and the specific application of the robot.

To facilitate more objective comparisons, we introduce the **Calibration Efficiency Index (CEI)**, defined as:

$$\text{CEI} = \frac{\omega_a A}{\omega_c C + \omega_s S + \omega_p P} \quad (9)$$

In this equation, A denotes the normalized accuracy improvement, C represents the cost score, S reflects the setup complexity, and P is the computational burden. The values for C , S , and P are rated on a scale from 1 to 5, with higher values indicating greater cost, complexity, and computational burden. The weights are assigned as: $[\omega_a \ \omega_c \ \omega_s \ \omega_p] = [5, 0.5, 0.35, 0.15]$.

The weights reflect the relative significance of each factor. The normalized accuracy improvement is assigned the highest weight, also serving to bound the CEI value between 0 and 5. Cost is the next most important factor, followed by setup complexity, which is influenced by factors such as the number of calibration points and the tools required. Computational burden is considered the least critical, especially for the offline calibration methods mainly addressed in this study.

Fig. 3 summarizes the methods reviewed in this paper, displaying the CEI metrics, normalized accuracy improvements, estimated cost, complexity, modeling, and optimization frameworks.

The proposed CEI offers a first step toward quantitative comparison. The use of specific weightings facilitates a semi-objective evaluation of the methods. However, while the CEI provides a useful comparison framework, it remains influenced by the chosen weightings and varying non-quantifiable experimental conditions across studies.

VII. UNSOLVED PROBLEMS AND FUTURE DIRECTIONS

- 1) **Mobile Manipulators:** Few calibration methods address mobile manipulators as integrated systems. Most approaches treat the mobile base and manipulator separately, neglecting coupling effects such as localization uncertainty, mounting errors, and compliance. Unified calibration frameworks enable joint modeling but introduce challenges in parameter identifiability and model complexity.
- 2) **Dynamic Parameter Calibration:** Current methods primarily focus on kinematic errors, while dynamic effects are often neglected. Factors such as compliance, backlash, thermal deformation, and payload-dependent deflections introduce configuration-dependent errors not captured by geometric models. Integrating geometric and dynamic calibration remains an open challenge.
- 3) **AI-Assisted Calibration:** Machine-learning approaches can model nonlinear error sources that are difficult to describe analytically. These approaches can learn relationships between joint configurations, payload conditions, and resulting tool center point behavior. However, challenges remain in generalization, integration with physics-based models, and ensuring robustness in safety-critical applications.
- 4) **Adaptive and Online Calibration:** Most calibration methods are performed offline, limiting applicability

in dynamic environments. Online approaches enable continuous parameter updates to compensate for wear and environmental changes, but require stable, efficient estimation methods and mechanisms to prevent drift and overfitting. Achieving robust real-time performance remains challenging.

VIII. CONCLUSION

This review presents a comprehensive overview of calibration methods for serial link manipulators, examining their evolution, key techniques, and implications for future research and applications. It addresses the stages of modeling, measurement, and optimization, and demonstrates that high accuracy can be achieved through cost-effective methods, thereby making precise calibration more accessible, particularly in industrial and service robotics.

We introduce the **Calibration Efficiency Index (CEI)**, a practical metric designed for systematic comparison across calibration methods, taking into account factors such as accuracy, cost, setup complexity, and computational burden. This framework provides a semi-objective evaluation, offering researchers a standardized means to compare available methods.

Despite the contributions made, several challenges persist. The lack of standardized datasets and evaluation metrics complicates objective comparison, with results being influenced by various extraneous factors.

Future research should prioritize directions such as mobile manipulators, dynamic parameters, machine-learning-based approaches, and adaptive or online calibration. Progress in these areas, coupled with the establishment of standardized methodologies, is critical for improving the overall effectiveness and precision of robot calibration techniques.

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REFERENCES

- [1] Y. Cong, R. Chen, B. Ma, H. Liu, D. Hou, and C. Yang, "A comprehensive study of 3-D vision-based robot manipulation," *IEEE Transactions on Cybernetics*, vol. 53, no. 3, pp. 1682–1698, 2021.
- [2] Z. Pan, Z. Jia, K. Jing, Y. Ding, and Q. Liang, "Manipulator Package Sorting and Placing System Based on Computer Vision," in *2020 Chinese Control And Decision Conference (CCDC)*, 2020, pp. 409–414.
- [3] Q. Liu, J. Mao, L. Han, C. Zhang, and J. Yang, "Predictive observer-based dual-rate prescribed performance control for visual servoing of robot manipulators with view constraints," *IEEE Transactions on Cybernetics*, 2025.
- [4] K. Li, H. Zhang, and H. Wang, "Nonlinear observer-based visual servoing and vibration control of flexible robotic manipulators with a fixed camera," *IEEE Transactions on Cybernetics*, vol. 54, no. 7, pp. 4100–4110, 2024.
- [5] Z. Roth, B. Mooring, and B. Ravani, "An overview of robot calibration," *IEEE Journal on Robotics and Automation*, vol. 3, no. 5, pp. 377–385, 1987.
- [6] A. Elatta, L. P. Gen, F. L. Zhi, Y. Daoyuan, and L. Fei, "An overview of robot calibration," *Information Technology Journal*, vol. 3, no. 1, pp. 74–78, 2004.
- [7] Z. Li, S. Li, and X. Luo, "An overview of calibration technology of industrial robots," *IEEE/CAA Journal of Automatica Sinica*, vol. 8, no. 1, pp. 23–36, 2021.
- [8] H. Cheng and K. C. Gupta, "An Historical Note on Finite Rotations," *Journal of Applied Mechanics*, vol. 56, no. 1, pp. 139–145, 03 1989. [Online]. Available: <https://doi.org/10.1115/1.3176034>
- [9] M. Chasles, "Note sur les propriétés générales du système de deux corps semblables entr'eux et placés d'une manière quelconque dans l'espace; et sur le déplacement fini ou infiniment petit d'un corps solide libre [A note on the general properties of a system of two similar bodies arbitrarily positioned in space; and on the finite or infinitely small displacement of an unconstrained solid body]," *Bulletin des Sciences Mathématiques, Férussac*, vol. 14, pp. 321–26, 1830.
- [10] W. R. Hamilton, *Elements of quaternions*. London: Longmans, Green, & Company, 1866.
- [11] Clifford, "Preliminary sketch of biquaternions," *Proceedings of the London Mathematical Society*, vol. 1, no. 1, pp. 381–395, 1871.
- [12] E. Study, "Die Geometrie der Dynamen," *Jahresbericht der Deutschen Mathematiker-Vereinigung*, vol. 8, pp. 204–216, 1900. [Online]. Available: <http://eudml.org/doc/144667>
- [13] M. Ulrich and C. Steger, "Hand-eye calibration of SCARA robots using dual quaternions," *Pattern Recognition and Image Analysis*, vol. 26, pp. 231–239, 2016.
- [14] Z. Fu, J. Pan, E. Spyrakos-Papastavridis, X. Chen, and M. Li, "A Dual Quaternion-Based Approach for Coordinate Calibration of Dual Robots in Collaborative Motion," *IEEE Robotics and Automation Letters*, vol. 5, no. 3, pp. 4086–4093, 2020.
- [15] *Plücker coordinates*. New York, NY: Springer New York, 2005, pp. 273–288. [Online]. Available: https://doi.org/10.1007/0-387-27103-1_14
- [16] R. S. Ball, *A Treatise on the Theory of Screws*. Cambridge university press, 1998.
- [17] B. C. Hall and B. C. Hall, *Lie groups, Lie algebras, and representations*. Springer, 2013.
- [18] R. W. Brockett, "Robotic manipulators and the product of exponentials formula," in *Mathematical Theory of Networks and Systems: Proceedings of the MTNS-83 International Symposium Beer Sheva, Israel, June 20–24, 1983*. Springer, 2005, pp. 120–129.
- [19] G. Chen, H. Wang, and Z. Lin, "Determination of the Identifiable Parameters in Robot Calibration based on the POE Formula," *IEEE Transactions on Robotics*, vol. 30, no. 5, pp. 1066–1077, 2014.
- [20] R. He, Y. Zhao, S. Yang, and S. Yang, "Kinematic-Parameter Identification for Serial-Robot Calibration Based on POE Formula," *IEEE Transactions on Robotics*, vol. 26, no. 3, pp. 411–423, 2010.
- [21] H. Wang, S. Shen, and X. Lu, "A screw axis identification method for serial robot calibration based on the POE model," *Industrial Robot: An International Journal*, vol. 39, no. 2, pp. 146–153, 2012.
- [22] I.-M. Chen, G. Yang, C. T. Tan, and S. H. Yeo, "Local POE model for robot kinematic calibration," *Mechanism and Machine Theory*, vol. 36, no. 11–12, pp. 1215–1239, 2001.
- [23] J. Denavit and R. S. Hartenberg, "A kinematic notation for lower-pair mechanisms based on matrices," 1955.
- [24] R. P. Paul, *Robot manipulators: mathematics, programming, and control: the computer control of robot manipulators*. Richard Paul, 1981.
- [25] S. Hayati and M. Mirmirani, "Improving the absolute positioning accuracy of robot manipulators," *Journal of Robotic Systems*, vol. 2, no. 4, pp. 397–413, 1985. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.4620020406>
- [26] H. Stone, A. Sanderson, and C. Neuman, "Arm signature identification," in *Proceedings. 1986 IEEE International Conference on Robotics and Automation*, vol. 3, 1986, pp. 41–48.
- [27] H. Zhuang, Z. S. Roth, and F. Hamano, "A complete and parametrically continuous kinematic model for robot manipulators," in *Proceedings., IEEE International Conference on Robotics and Automation*. IEEE, 1990, pp. 92–97.

- [28] M. Theofanidis, S. I. Sayed, J. Cloud, J. Brady, and F. Makedon, "Kinematic estimation with neural networks for robotic manipulators," in *Artificial Neural Networks and Machine Learning-ICANN 2018: 27th International Conference on Artificial Neural Networks, Rhodes, Greece, October 4-7, 2018, Proceedings, Part III* 27. Springer, 2018, pp. 795–802.
- [29] B. Daya, S. Khawandi, M. Akoum *et al.*, "Applying neural network architecture for inverse kinematics problem in robotics," *Journal of Software Engineering and Applications*, vol. 3, no. 03, p. 230, 2010.
- [30] R. Köker, C. Öz, T. Çakar, and H. Ekiz, "A study of neural network based inverse kinematics solution for a three-joint robot," *Robotics and autonomous systems*, vol. 49, no. 3-4, pp. 227–234, 2004.
- [31] R. Köker, "A genetic algorithm approach to a neural-network-based inverse kinematics solution of robotic manipulators based on error minimization," *Information Sciences*, vol. 222, pp. 528–543, 2013.
- [32] J. Demby's, Y. Gao, and G. N. DeSouza, "A study on solving the inverse kinematics of serial robots using artificial neural network and fuzzy neural network," in *2019 IEEE international conference on fuzzy systems (FUZZ-IEEE)*. IEEE, 2019, pp. 1–6.
- [33] Stanford Artificial Intelligence Laboratory *et al.*, "Robotic Operating System," <https://www.ros.org>.
- [34] B. Yu, J. Lee, S. H. Park, and M. Kim, "From Unified Robot Description Format to DH Parameters: Examinations of Two Different Approaches for Manipulator," *IEEE Access*, vol. 12, pp. 133 441–133 455, 2024.
- [35] J. He, M. Zhang, F. Xu, Q. Liu, and H. Li, "A Model Conversion Algorithm from URDF to DH for Camera Robot," in *2022 International Conference on Culture-Oriented Science and Technology (CoST)*, 2022, pp. 134–139.
- [36] R. Hartenberg and J. Denavit, *Kinematic synthesis of linkages*. New York: McGraw-Hill, 1964.
- [37] C. Faria, J. L. Vilaça, S. Monteiro, W. Erhagen, and E. Bicho, "Automatic Denavit-Hartenberg Parameter Identification for Serial Manipulators," in *IECON 2019 - 45th Annual Conference of the IEEE Industrial Electronics Society*, vol. 1, 2019, pp. 610–617.
- [38] K.-Y. Young, J.-J. Chen, and C.-C. Wang, "An automated robot calibration system based on a variable d-h parameter model," in *Proceedings of 35th IEEE Conference on Decision and Control*, vol. 1, 1996, pp. 881–886 vol.1.
- [39] J.-W. Lee, G.-T. Park, J.-S. Shin, and J.-W. Woo, "Industrial robot calibration method using denavit — hatenberg parameters," in *2017 17th International Conference on Control, Automation and Systems (ICCAS)*, 2017, pp. 1834–1837.
- [40] L. K. Barker, "Modified Denavit-Hartenberg parameters for better location of joint axis systems in robot arms," *Tech. Rep.*, 1986.
- [41] H. Wang, H. Qi, M. Xu, Y. Tang, J. Yao, X. Yan, and M. Li, "Research on the Relationship between Classic Denavit-Hartenberg and Modified Denavit-Hartenberg," in *2014 Seventh International Symposium on Computational Intelligence and Design*, vol. 2, 2014, pp. 26–29.
- [42] S. Hayati, K. Tso, and G. Roston, "Robot geometry calibration," in *Proceedings. 1988 IEEE International Conference on Robotics and Automation*, 1988, pp. 947–951 vol.2.
- [43] N. Juneja and A. A. Goldenberg, "Kinematic calibration of a re-configurable robot (robotwin)," in *Proceedings of International Conference on Robotics and Automation*, vol. 4. IEEE, 1997, pp. 3178–3183.
- [44] L. S. Ginani and J. M. S. Motta, "Theoretical and practical aspects of robot calibration with experimental verification," *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, vol. 33, pp. 15–21, 2011.
- [45] P. Rousseau, A. Desrochers, and N. Krouglicof, "Machine vision system for the automatic identification of robot kinematic parameters," *IEEE Transactions on Robotics and Automation*, vol. 17, no. 6, pp. 972–978, 2001.
- [46] J. Kuti and P. Galambos, *Conversion Between Denavit-Hartenberg and Hayati Conventions for Calibration and Forward Kinematics of Manipulators*. Cham: Springer Nature Switzerland, 2024, pp. 35–47. [Online]. Available: https://doi.org/10.1007/978-3-031-58257-8_3
- [47] C. Rocha, C. Tonetto, and A. Dias, "A comparison between the Denavit-Hartenberg and the screw-based methods used in kinematic modeling of robot manipulators," *Robotics and Computer-Integrated Manufacturing*, vol. 27, no. 4, pp. 723–728, 2011.
- [48] L. J. Everett and T.-W. Hsu, "The theory of kinematic parameter identification for industrial robots," 1988.
- [49] H. Zhuang and Z. S. Roth, "Robot calibration using the cpc error model," *Robotics and computer-integrated manufacturing*, vol. 9, no. 3, pp. 227–237, 1992.
- [50] H. Zhuang, L. K. Wang, and Z. S. Roth, "Error-model-based robot calibration using a modified cpc model," *Robotics and computer-integrated manufacturing*, vol. 10, no. 4, pp. 287–299, 1993.
- [51] H. Zhuang and Z. S. Roth, "A linear solution to the kinematic parameter identification of robot manipulators," *IEEE Transactions on Robotics and Automation*, vol. 9, no. 2, pp. 174–185, 1993.
- [52] K. Lynch, *Modern Robotics*. Cambridge University Press, 2017.
- [53] R. Mukundan and R. Mukundan, "Quaternions," *Advanced Methods in Computer Graphics: With examples in OpenGL*, pp. 77–112, 2012.
- [54] L. Kavan, S. Collins, C. O'Sullivan, and J. Zara, "Dual quaternions for rigid transformation blending," *Trinity College Dublin*, vol. 46, 2006.
- [55] B. V. Adorno, "Robot Kinematic Modeling and Control Based on Dual Quaternion Algebra—Part i: Fundamentals," 2017.
- [56] G. Leclercq, P. Lefèvre, and G. Blohm, "3d kinematics using dual quaternions: theory and applications in neuroscience," *Frontiers in behavioral neuroscience*, vol. 7, p. 7, 2013.
- [57] J. Angeles, "The application of dual algebra to kinematic analysis," *Computational Methods in Mechanical Systems: Mechanism Analysis, Synthesis, and Optimization*, pp. 3–32, 1998.
- [58] Y.-L. Gu and J. Luh, "Dual-number transformation and its applications to robotics," *IEEE Journal on Robotics and Automation*, vol. 3, no. 6, pp. 615–623, 1987.
- [59] J. Voight, *Quaternion algebras*. Springer Nature, 2021.
- [60] F. Zhang, "Quaternions and matrices of quaternions," *Linear algebra and its applications*, vol. 251, pp. 21–57, 1997.
- [61] C. Pozna and R.-E. Precup, "On the Use of Quaternions, in the Translated Reference Frame Formalism," *Acta Polytechnica Hungarica*, vol. 20, no. 6, pp. 195–214, 2023.
- [62] Y.-B. Jia, "Dual quaternions," *Iowa State University: Ames, IA, USA*, 2013.
- [63] B. Kenwright, "A beginners guide to dual-quaternions: what they are, how they work, and how to use them for 3d character hierarchies," 2012.
- [64] W. Khalil, M. Gautier, and C. Enguehard, "Identifiable parameters and optimum configurations for robots calibration," *Robotica*, vol. 9, no. 1, pp. 63–70, 1991.
- [65] T. Li, K. Sun, Z.-w. Xie, and H. Liu, "Optimal measurement configurations for kinematic calibration of six-dof serial robot," *Journal of Central South University*, vol. 18, no. 3, pp. 618–626, 2011.
- [66] Z. Li, S. Li, and X. Luo, "Efficient industrial robot calibration via a novel unscented Kalman filter-incorporated variable step-size Levenberg-Marquardt algorithm," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–12, 2023.
- [67] Z. Li, S. Li, and H. Wu, "A new variable step-size levenberg-marquardt algorithm for industrial robot calibration," in *2022 IEEE international conference on networking, sensing and control (ICNSC)*. IEEE, 2022, pp. 1–6.
- [68] T. Chen, S. Li, Y. Qiao, and X. Luo, "A robust and efficient ensemble of diversified evolutionary computing algorithms for accurate robot calibration," *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–14, 2024.
- [69] G. Boschetti and T. Sinico, "A novel step-by-step procedure for the kinematic calibration of robots using a single draw-wire encoder," *The International Journal of Advanced Manufacturing Technology*, vol. 131, no. 7, pp. 4129–4147, 2024.
- [70] S. He, L. Ma, C. Yan, C.-H. Lee, and P. Hu, "Multiple location constraints based industrial robot kinematic parameter calibration and accuracy assessment," *The International Journal of Advanced Manufacturing Technology*, vol. 102, pp. 1037–1050, 2019.
- [71] W. Veitschegger and C.-h. Wu, "A method for calibrating and compensating robot kinematic errors," in *Proceedings. 1987 IEEE International Conference on Robotics and Automation*, vol. 4. IEEE, 1987, pp. 39–44.
- [72] J. Liu, Y. Zhang, and Z. Li, "Improving the Positioning Accuracy of a Neurosurgical Robot System," *IEEE/ASME Transactions on Mechatronics*, vol. 12, no. 5, pp. 527–533, 2007.
- [73] H. Zhuang, S. H. Motaghedi, and Z. S. Roth, "Robot calibration with planar constraints," in *Proceedings 1999 IEEE International Conference on Robotics and Automation (Cat. No. 99CH36288C)*, vol. 1. IEEE, 1999, pp. 805–810.
- [74] S. Besnard, W. Khalil, and G. Garcia, "Geometric calibration of robots using multiple plane constraints," in *Advances in robot kinematics*. Springer, 2000, pp. 61–70.
- [75] O. Sumenkov, "Practical aspects of automated manipulator calibration with plane constraint," in *2025 International Conference on Industrial Engineering, Applications and Manufacturing (ICIEAM)*, 2025, pp. 924–929.

- [76] X.-L. Zhong and J. M. Lewis, "A new method for autonomous robot calibration," in *Proceedings of 1995 IEEE international conference on robotics and automation*, vol. 2. IEEE, 1995, pp. 1790–1795.
- [77] M. Driels, "Using passive end-point motion constraints to calibrate robot manipulators," 1993.
- [78] A. Joubair and I. A. Bonev, "Kinematic calibration of a six-axis serial robot using distance and sphere constraints," *The International Journal of Advanced Manufacturing Technology*, vol. 77, pp. 515–523, 2015.
- [79] A. Nubiola and I. A. Bonev, "Absolute robot calibration with a single telescoping ballbar," *Precision Engineering*, vol. 38, no. 3, pp. 472–480, 2014.
- [80] N. A. Nadeau, I. A. Bonev, and A. Joubair, "Impedance control self-calibration of a collaborative robot using kinematic coupling," *Robotics*, vol. 8, no. 2, p. 3, 2019.
- [81] T. Messay, R. Ordóñez, and E. Marcil, "Computationally efficient and robust kinematic calibration methodologies and their application to industrial robots," *Robotics and Computer-Integrated Manufacturing*, vol. 37, pp. 33–48, 2016.
- [82] Y. Gan, J. Duan, and X. Dai, "A calibration method of robot kinematic parameters by drawstring displacement sensor," *International Journal of Advanced Robotic Systems*, vol. 16, no. 5, 2019.
- [83] R. Wang, A. Wu, X. Chen, and J. Wang, "A point and distance constraint based 6r robot calibration method through machine vision," *Robotics and Computer-Integrated Manufacturing*, vol. 65, 2020.
- [84] M. Švaco, B. Šekoranja, F. Šuligoj, and B. Jerbić, "Calibration of an industrial robot using a stereo vision system," *Procedia Engineering*, vol. 69, pp. 459–463, 2014.
- [85] J. Misekikis, K. Glette, O. J. Elle, and J. Torresen, "Automatic calibration of a robot manipulator and multi 3d camera system," in *2016 IEEE/SICE International Symposium on System Integration (SII)*. IEEE, 2016, pp. 735–741.
- [86] R. A. Boby, "Kinematic Identification of Industrial Robot Using End-Effector Mounted Monocular Camera Bypassing Measurement of 3-d Pose," *IEEE/ASME Transactions on Mechatronics*, vol. 27, no. 1, pp. 383–394, 2022.
- [87] O. Kroeger, J. Huegle, and C. A. Niebuhr, "An automatic calibration approach for a multi-camera-robot system," in *2019 24th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, 2019, pp. 1515–1518.
- [88] I.-W. Park, B.-J. Lee, S.-H. Cho, Y.-D. Hong, and J.-H. Kim, "Laser-Based Kinematic Calibration of Robot Manipulator Using Differential Kinematics," *IEEE/ASME Transactions on Mechatronics*, vol. 17, no. 6, pp. 1059–1067, 2012.
- [89] H.-N. Nguyen, J. Zhou, and H.-J. Kang, "A calibration method for enhancing robot accuracy through integration of an extended Kalman filter algorithm and an artificial neural network," *Neurocomputing*, vol. 151, pp. 996–1005, 2015.
- [90] T. Chen, J. Lin, D. Wu, and H. Wu, "Research of calibration method for industrial robot based on error model of position," *Applied Sciences*, vol. 11, no. 3, 2021.
- [91] A. Kohama, R. Mori, S. Komai, M. Suzuki, S. Aoyagi, J. Fujioka, and Y. Kamiya, "Calibration of kinematic parameters of a robot using neural networks by a laser tracking system," in *Service Robotics and Mechatronics: Selected Papers of the International Conference on Machine Automation ICMA2008*. Springer, 2010, pp. 269–274.
- [92] L. Ma, P. Bazzoli, P. M. Sammons, R. G. Landers, and D. A. Bristow, "Modeling and calibration of high-order joint-dependent kinematic errors for industrial robots," *Robotics and Computer-Integrated Manufacturing*, vol. 50, pp. 153–167, 2018.
- [93] Z. Jiang, W. Zhou, H. Li, Y. Mo, W. Ni, and Q. Huang, "A new kind of accurate calibration method for robotic kinematic parameters based on the extended kalman and particle filter algorithm," *IEEE Transactions on Industrial Electronics*, vol. 65, no. 4, pp. 3337–3345, 2017.
- [94] J. Zhou and H.-J. Kang, "A hybrid least-squares genetic algorithm-based algorithm for simultaneous identification of geometric and compliance errors in industrial robots," *Advances in Mechanical Engineering*, vol. 7, no. 6, 2015.
- [95] J. Li, A. Ito, and Y. Maeda, "A SLAM-Integrated Kinematic Calibration Method for Industrial Manipulators with RGB-D Cameras," in *2019 19th International Conference on Control, Automation and Systems (ICCAS)*, 2019, pp. 686–689.
- [96] J. Li, A. Ito, H. Yaguchi, and Y. Maeda, "Simultaneous kinematic calibration, localization, and mapping (SKCLAM) for industrial robot manipulators," *Advanced Robotics*, vol. 33, no. 23, pp. 1225–1234, 2019.
- [97] H. G. P. Rune SØE-KNUDSEN, Esben Hallundbæk Østergaard, "Calibration and programming of robots, u.s. patent 9248573b2," <https://patents.google.com/patent/US9248573B2/en?q=calibration&assignee=Universal+Robots>.
- [98] F. Thomas, E. Ottaviano, L. Ros, and M. Ceccarelli, "Performance analysis of a 3-2-1 pose estimation device," *IEEE Transactions on Robotics*, vol. 21, no. 3, pp. 288–297, 2005.
- [99] J. W. Jeong, S. H. Kim, Y. K. Kwak, and C. C. Smith, "Development of a parallel wire mechanism for measuring position and orientation of a robot end-effector," *Mechatronics*, vol. 8, no. 8, pp. 845–861, 1998.
- [100] G. Legnani and M. Tiboni, "Optimal design and application of a low-cost wire-sensor system for the kinematic calibration of industrial manipulators," *Mechanism and Machine Theory*, vol. 73, pp. 25–48, 2014.
- [101] Y. Liu, Z. Zhuang, and Y. Li, "Closed-Loop Kinematic Calibration of Robots Using a Six-Point Measuring Device," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–12, 2022.
- [102] L. Yang, Q. Cao, M. Lin, H. Zhang, and Z. Ma, "Robotic hand-eye calibration with depth camera: A sphere model approach," in *2018 4th International Conference on Control, Automation and Robotics (ICCAR)*, 2018, pp. 104–110.
- [103] Y. Meng and H. Zhuang, "Autonomous robot calibration using vision technology," *Robotics and Computer-Integrated Manufacturing*, vol. 23, no. 4, pp. 436–446, 2007.
- [104] J. M. S. Motta, G. C. De Carvalho, and R. McMaster, "Robot calibration using a 3d vision-based measurement system with a single camera," *Robotics and Computer-Integrated Manufacturing*, vol. 17, no. 6, pp. 487–497, 2001.
- [105] G. Du and P. Zhang, "Online robot calibration based on vision measurement," *Robotics and Computer-Integrated Manufacturing*, vol. 29, no. 6, pp. 484–492, 2013.
- [106] Q. Zhu, X. Xie, C. Li, G. Xia, and Q. Liu, "Kinematic Self-Calibration Method for Dual-Manipulators Based on Optical Axis Constraint," *IEEE Access*, vol. 7, pp. 7768–7782, 2019.
- [107] H. M. Balanji, A. E. Turgut, and L. T. Tunc, "A novel vision-based calibration framework for industrial robotic manipulators," *Robotics and Computer-Integrated Manufacturing*, vol. 73, 2022.
- [108] X. Zhang, Y. Song, Y. Yang, and H. Pan, "Stereo vision based autonomous robot calibration," *Robotics and Autonomous Systems*, vol. 93, pp. 43–51, 2017.
- [109] G. Du, P. Zhang, and D. Li, "Online robot calibration based on hybrid sensors using kalman filters," *Robotics and Computer-Integrated Manufacturing*, vol. 31, pp. 91–100, 2015.
- [110] A. Nubiola and I. A. Bonev, "Absolute calibration of an ABB IRB 1600 robot using a laser tracker," *Robotics and Computer-Integrated Manufacturing*, vol. 29, no. 1, pp. 236–245, 2013.
- [111] A. Filon, A. Joubair, A. S. Tahan, and I. A. Bonev, "Robot calibration using a portable photogrammetry system," *Robotics and Computer-Integrated Manufacturing*, vol. 49, pp. 77–87, 2018.
- [112] B. Dasgupta and T. Mruthyunjaya, "The Stewart platform manipulator: a review," *Mechanism and Machine Theory*, vol. 35, no. 1, pp. 15–40, 2000. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0094114X99000063>
- [113] H. Abdi *et al.*, "The method of least squares," *Encyclopedia of measurement and statistics*, vol. 1, pp. 530–532, 2007.
- [114] Y. Wang, "Gauss-newton method," *Wiley Interdisciplinary Reviews: Computational Statistics*, vol. 4, no. 4, pp. 415–420, 2012.
- [115] S. Ruder, "An overview of gradient descent optimization algorithms," *arXiv preprint arXiv:1609.04747*, 2016.
- [116] Y. Kim, H. Bang *et al.*, "Introduction to Kalman filter and its applications," *Introduction and Implementations of the Kalman Filter*, vol. 1, pp. 1–16, 2018.
- [117] E. A. Wan and R. Van Der Merwe, "The unscented Kalman filter," *Kalman filtering and neural networks*, pp. 221–280, 2001.
- [118] P. M. Djuric, J. H. Kotecha, J. Zhang, Y. Huang, T. Ghirmai, M. F. Bugallo, and J. Miguez, "Particle filtering," *IEEE signal processing magazine*, vol. 20, no. 5, pp. 19–38, 2003.
- [119] I. J. Myung, "Tutorial on maximum likelihood estimation," *Journal of mathematical Psychology*, vol. 47, no. 1, pp. 90–100, 2003.
- [120] S. Singer and J. Nelder, "Nelder-mead algorithm," *Scholarpedia*, vol. 4, no. 7, p. 2928, 2009.
- [121] A. E. Eiben and J. E. Smith, *Introduction to evolutionary computing*. Springer, 2015.
- [122] K. Wang, "Application of genetic algorithms to robot kinematics calibration," *International Journal of Systems Science*, vol. 40, no. 2, pp. 147–153, 2009.
- [123] Y. Liu, B. Liang, W. Qiang, and J. Yan-shu, "Improvement on robots positioning accuracy based on genetic algorithm," in *The Proceedings*

of the Multiconference on" Computational Engineering in Systems Applications", vol. 1. IEEE, 2006, pp. 387–392.

- [124] J. H. Holland, "Genetic algorithms," *Scientific american*, vol. 267, no. 1, pp. 66–73, 1992.
- [125] S. Sivanandam, S. Deepa, S. Sivanandam, and S. Deepa, *Genetic algorithms*. Springer, 2008.

BIOGRAPHY SECTION



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